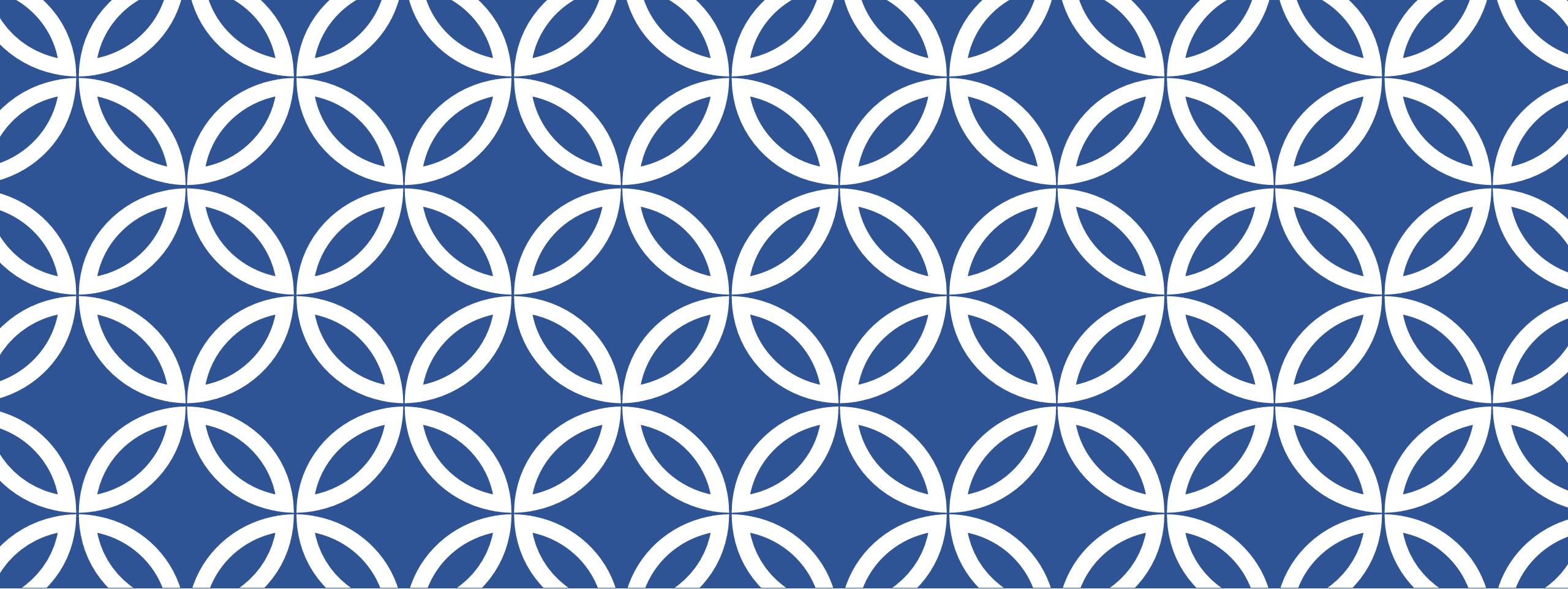




Satellite Internet communication using the osi layer model

DONE BY

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What Is Satellite Internet?

Delivering connectivity from orbit, layer by layer

- Delivers connectivity via space-based relays, reaching areas fibre and cell towers can't.
- LEO, MEO & GEO satellites trade altitude for latency and coverage.
- The OSI model still governs the link — every layer must hold up across thousands of km of vacuum.

550 km

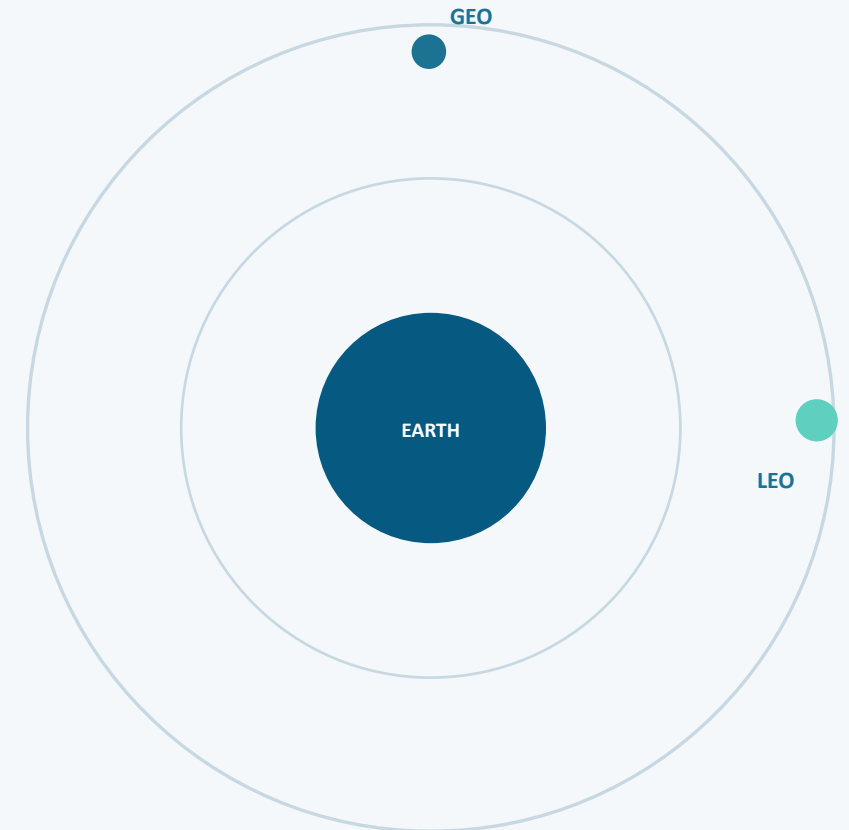
Typical LEO
altitude

~25 ms

Round-trip
latency (LEO)

35,786 km

GEO orbital
altitude



Mapping the OSI Stack to Satellite Links

How each layer stays reliable across a space-based hop

7	Application	Web, video & VoIP traffic over the satellite link
6	Presentation	Compression & encryption to offset limited bandwidth
5	Session	Keeps sessions alive through satellite hand-offs
4	Transport	TCP tuned for high latency; fewer retransmits
3	Network	IP routing via inter-satellite links & gateways
2	Data Link	Error control for signal fade & Doppler shift
1	Physical	Ku/Ka-band RF signal, earth-to-orbit

Advantages & Challenges

Reliability trade-offs unique to a space-based OSI path

ADVANTAGES

- Reaches remote areas with no fibre or cell towers
- Rapid deployment — no ground trenching or cabling
- LEO now delivers broadband-comparable speeds
- Resilient when regional disasters sever ground links

CHALLENGES

- GEO delay can exceed 500 ms round-trip
- Rain fade & atmospheric signal attenuation
- LEO hand-offs can drop packets in transit
- Higher cost per subscriber than fibre

Conclusion & Future Scope

Beyond the traditional OSI model

- OSI stays the shared language for reliable transmission — even over a radio link to orbit.
- LEO mega-constellations push tighter Network–Transport integration for real-time routing.
- Next up: AI traffic management, laser inter-satellite links, hybrid terrestrial handovers.

Key Takeaway: Reliable networking is layer-independent — satellite internet proves the OSI model scales from copper wire to low Earth orbit.